

Leif Adamec Rydenfalk

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SUMMARY

Systems engineer and founder who builds the systems companies usually rent — and operates them in production. Currently running a peer-to-peer compute mesh with its own credit economy (70+ services, live paid workloads), a resale marketplace with full payment flows, distributed LLM inference across a heterogeneous fleet, and a multi-platform AI SaaS. Ships 500k+ lines of production Rust/TypeScript a year solo by engineering the development process itself around fleets of AI agents. Self-taught; learns from first principles and builds the whole stack — from sub-microsecond IPC to payment settlement to the deploy pipeline.

EXPERIENCE

Systems Engineer – Internal AI Tools | Gaming Studio, Umeå

March 2025 – Present

- Built an autonomous "time-guardian" in Rust (Axum/Tokio): ingests employee time reports from Fortnox, analyzes them against company policy with an LLM, and nudges employees over Slack — payroll reconciliation went from ~3 days to ~30 minutes.
- Runs unattended in production: automatic OAuth token refresh, autonomous Fortnox↔Slack identity mapping, admin dashboard behind Google OAuth, zero-downtime deploys over Cloudflare Tunnels.
- Property-based tests on the validation engine caught 11 edge-case bugs before production.

Founder / Systems Engineer | Rydenfalk Systems, Umeå

February 2025 – Present

- **CE / ce-net** (ce-net.com) — designed, built, and operate a peer-to-peer compute mesh and economy: run a node, donate compute, earn credits, spend them on other people's machines. Ed25519 identity, signed capability chains as the only trust primitive (attenuation, revocation), a PoW credit ledger with escrow and payment channels, gVisor-sandboxed workloads. The paid compute economy is live — usage metered, admitted, and settled across a multi-machine fleet (Hetzner relay, GPU workstation, laptops, in-browser WASM nodes). Packaged for Homebrew, AUR, Scoop, Chocolatey.
- Built **70+ services on the mesh** rebuilding Google-class infrastructure as apps over one SDK: S3-class object storage, Firestore-class realtime document DB, GKE-style orchestration, pub/sub with durable replay, CDN, serverless functions, IAM with passwordless device pairing, distributed CI, capability-gated tunnels.
- **Distributed LLM inference** — ce-exo serves large models across the heterogeneous fleet by wrapping production engines (vLLM, llama.cpp, exo, Ollama) behind one OpenAI-compatible API with NAT-traversing transport and capability-gated access; ce-tabnet shards a transformer layer-by-layer across browser tabs (pipeline-parallel WebGPU inference). Designed a clinical deployment where inference runs on hospital GPUs and PHI never leaves the LAN.
- **Myra / Trana** (trana.app) — AI lesson-material SaaS for Swedish teachers. One TypeScript monorepo ships four clients (web, mobile via App Store/Play Store, desktop, landing) from shared packages; Supabase with row-level security; full test pyramid in per-app CI/CD.
- **Blueberry Browser** — Electron/Chromium browser exposing an MCP server so AI agents operate a real, logged-in session; automates Gmail, WhatsApp Web, LinkedIn with no per-site integrations.
- Engineer the development process around fleets of parallel AI agents — recorded conventions, shared findings backlogs, remote build offloading, reproduce-first bug methodology — sustaining 500k+ production lines a year, solo. Foundation: **Cell**, a zero-copy Rust substrate at 1.48M msg/s (~677ns median RTT, single core) with embeddable Raft.

Full-Stack Contractor | Self Employed, Umeå

January 2024 – May 2025

- Delivered commercial codebases (SvelteKit + Node.js + PostgreSQL) with CI/CD via GitHub Actions.
- Integrated Stripe Connect split payments; handled PCI-sensitive flows via Stripe webhooks.
- Containerized all services with Docker; dev-prod parity under a 2-minute cold start.

SELECTED PROJECTS

CE / ce-net — ce-net.com · P2P compute mesh and economy in Rust: libp2p (Kademlia, Gossipsub, relays, hole punching), PoW credit ledger with integer base-unit money, escrowed job settlement, payment channels with off-chain receipts, capability-chain IAM, gVisor sandboxing. 70+ services; live paid economy on a real fleet.

ce-exo + ce-tabnet — github.com/ce-net/ce-exo · Distributed LLM inference: one OpenAI-compatible API over a NAT-traversing mesh wrapping vLLM / llama.cpp / exo / Ollama — one command deploys an engine to a GPU machine and routes chat traffic to it, capability-authed. Plus a transformer sharded layer-by-layer across browser tabs (pipeline-parallel WebGPU).

Rheo — rheo.se · Mobile-first resale marketplace in Rust: Axum, React Native, PostgreSQL, Redis Streams, Stripe Connect (onboarding, escrowed split payments), PostNord shipping, R2. Multi-currency with per-listing anchor currency. Three fail-independent monitoring layers; p95 ~78ms on a 256MB container; blue-green deploys.

Myra / Trana — trana.app · AI lesson-material SaaS: one TypeScript monorepo → web (TanStack Start + React 19 on Cloudflare Workers), landing (Astro), mobile (Expo), desktop (Electron). Supabase (RLS); Vitest, Playwright, pgTAP; per-app CI/CD.

Spacegame — github.com/ce-net/spacegame · Browser multiplayer with no game server: each browser runs the full authoritative simulation (Rust → WASM + WebGL2), peers merge state by quorum — replicated authority. Zero-delay prediction, per-RTT update rates, population-driven persistence.

Also: Blueberry Browser (MCP agent browser), Voxel game engine (Rust/wgpu, 60 FPS on A15/M1), Cell (1.48M msg/s substrate), OpenJaws (typed cell mesh), Cerena (10k-player distributed RPG, in progress), ce-cast (distributed live-streaming studio), Dream Engine, Mariana.

SKILLS

- **Languages:** Rust (advanced), TypeScript, JavaScript, Python, SQL, WGSL, C, C++
- **Distributed systems:** libp2p (DHT, gossip, NAT traversal), PoW ledgers, Raft, capability security (attenuation/revocation), zero-copy IPC (rkyv), shared-memory transports, BFT design, Sybil resistance
- **LLM / AI infrastructure:** distributed inference across heterogeneous fleets, pipeline parallelism, WebGPU inference, quantized serving (GGUF), vLLM, llama.cpp, exo, Ollama, OpenAI-compatible API design, LLM tool-use in production, MCP, multi-agent engineering workflows
- **Financial systems:** Stripe Connect (onboarding, escrow, split payouts, webhooks), credit ledger design (integer base units, supply caps, halving), payment channels and off-chain settlement, usage metering and admission control, multi-currency pricing
- **Backend:** Axum, Tokio, Node.js, REST/WebSockets, microservices, Redis Streams, PostgreSQL, Supabase
- **Frontend:** SvelteKit, React/React 19, React Native (Expo), TanStack Start, Astro, Electron, Tailwind CSS
- **Graphics:** wgpu, WGSL, SDF/ray-marching, voxel traversal, ECS (hecs), TAA, bloom, render graphs
- **Infrastructure & ops:** Cloudflare (Workers, Pages, Tunnels, R2, DNS), Hetzner, Docker, gVisor, systemd fleets, blue-green deploys, deploy smoke gates, fail-independent monitoring, GitHub Actions, cross-platform release CI, package distribution (Homebrew, AUR, Scoop, Chocolatey)
- **Testing:** TDD, property-based testing, reproduce-first bug methodology, Playwright/Vitest E2E, pgTAP, live multi-node e2e harnesses

EDUCATION

Self-taught.